Probability Homework #3

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1. In a closet there are 10 pairs of shoes. If six shoes are selected at random, what is the probability of (a) no complete pairs; (b) exactly one complete pair?.

(a) $N(S) = C_6^{20} = 38760$

$N(A) = C_6^{10} \times 2^6 = 13440$

$P(A) = \frac{13440}{38760} \approx 34.675\%$

Ans: 34.675%

(b) $N(S) = C_6^{20} = 38760$

$N(A) = C_1^{10} \times C_4^9 \times 2^4 = 20160$

$P(A) = \frac{20160}{38760} \approx 52.012\%$

Ans: 52.012%

2. What is the coefficient of $x^2y^3z^2$ in the expansion of $(2x - y + 3z)^7$?

$\frac{7!}{2!3!2!} \times (2x)^2 \times (-y)^3 \times (3z)^2 = 210 \times 4x^2 \times -y^3 \times 9z^2 = -7560x^2y^3z^2$

Ans: -7560

3. A team consisting of three boys and four girls must be formed from a group of nine boys and eight girls. If two of the girls are feuding and refuse to play on the same team, how many possibilities do we have?

$N(S) = C_3^9 \times C_4^8 = 5880$

$N(A^C) = C_3^9 \times C_2^6 = 1260$

$N(A) = 5880 - 1260 = 4620$

Ans: 4620

4. A fair die is tossed six times. What is the probability of getting exactly two 6's?

$N(S) = 6^6 = 46656$

$N(A) = C_2^6 \times 5^4 = 9375$

$P(A) = \frac{9375}{46656} \approx 20.094\%$

Ans: 20.094%

5. By a combinatorial argument, prove that for $r \le n$ and $r \le m$,

$$\binom{n+m}{r} = \binom{m}{0}\binom{n}{r} + \binom{m}{1}\binom{n}{r-1} + \cdots + \binom{m}{r}\binom{n}{0}$$

1. 設 m 為班上的男生， n 為班上的女生，且 m+n 為班上的所有人數
2. 等式左側：從班上（n+m人）中選出 r 個人的可能性
3. 等式右側：
    a. 先從 m 中選取 0 人、再從 n 中選取 r-0 人
    b. 先從 m 中選取 1 人、再從 n 中選取 r-1 人
    c. 以此類推直到 從 m 中選取 r 人、n 中選取 r-r 人
    d. 由於每個步驟的男女人數都不一樣，所以他們都是 mutually exclusive，可以使用結合律相加
4. 由於右側等式也涵蓋了班上選 r 人的所有可能，因此左右策邏輯一致，為相等的式子

6. There are 12 balls in a box. Among these balls, two balls are yellow. Someone withdrew 7 balls randomly from the box. What is the probability that the remaining 5 balls contains at least 1 yellow ball?

$N(S) = C_7^{12} = 792$

$N(A^C) = C_2^2 \times C_5^{10} = 252$

$P(A) = 1 - \frac{252}{792} \approx 68.182\%$ Ans: 68.182 %

7. If eight defective and 12 nondefective items are inspected one by one, at random and without replacement, what is the probability that (a) the first four items inspected are defective; (b) from the first three items at least two are defective?

(a) $N(S) = C_4^{20} = 4845$

$N(A) = C_4^8 = 70$

$P(A) = \frac{70}{4845} \approx 1.445\%$ Ans: 1.445 %

(b) $N(S) = C_3^{20} = 1140$

$N(A) = C_2^8 \times C_1^{12} + C_3^8 \times C_0^{12} = 336 + 56 = 392$

$P(A) = \frac{392}{1140} \approx 34.386\%$ Ans: 34.386 %

8. From an ordinary deck of 52 cards, cards are drawn one by one, at random and without replacement. What is the probability that the fourth heart is drawn on the tenth draw? *Hint: Let F denote the event that in the first nine draws there are exactly three hearts, and E be the event that the tenth draw is a heart. Use P(FE) =P(F)P(E | F)*

Event F:

$$N(S) = C_9^{52} = \frac{52!}{9!43!} = 3679075400$$

$$N(F) = C_3^{13} \times C_6^{39} = \frac{13!}{10!3!} \times \frac{39!}{6!33!} = 933110178$$

Event E | F :

$$P(E|F) = \frac{10}{43}$$

$$P(FE) = \frac{933110178}{3679075400} \times \frac{10}{43} \approx 5.898\%$$

Ans: 5.898 %

9. Suppose that 40% of the students of a campus are men. If 20% of the men and 16% of the women of this campus own driver licenses, what percent of all of them own driver licenses?

Let P(A) be the probability that he is a men $P(A) = 40\%$

Let P(A^C) be the probability that she is a woman $P(A^C) = 60\%$

Let P(C) be the probability that he or she has driver licence

$P(C|A) = 20\%$ $\quad P(CA) = P(C|A) \times P(A) = 20\% \times 40\% = 8\%$

$P(C|A^C) = 16\%$ $\quad P(CA^C) = P(C|A^C) \times P(A^C) = 16\% \times 60\% = 9.6\%$

$$P(C) = P(AC^C) + P(AC) = 8\% + 9.6\% = 17.6\%$$

Ans: 17.6%

10. One of the cards of an ordinary deck of 52 cards is lost. What is the probability that a random card drawn from this deck is a spade?

Let A be a event that the lost card is a spade

Let B be a event that draw a card from a lost-1-card deck

$$P(A) = \frac{13}{52} \qquad P(A^C) = \frac{39}{52}$$

$$P(B|A) = \frac{12}{51} \qquad P(B|A^C) = \frac{13}{51}$$

$$P(BA) = P(B|A) \times P(A) = \frac{13 \times 12}{52 \times 51} \qquad P(BA^C) = P(B|A^C) \times P(A^C) = \frac{39 \times 13}{52 \times 51}$$

$$P(B) = P(BA) + P(BA^C) = \frac{663}{2652} = 25\%$$

Ans: 25%

11. A box contains 18 tennis balls, of which eight are new. Suppose that three balls are selected randomly, played with, and after play are returned to the box. If another three balls are selected for play a second time, what is the probability that they are all new?

Let event A be select 0 old ball and 3 new balls

Let event B be select 1 old ball and 2 new balls

Let event C be select 2 old balls and 1 new ball

Let event D be select 3 old balls and 0 new ball

Let event Z be selected 3 new balls from the box

$N(S) = C_3^{18} = 816$

$P(A) = \frac{C_3^8 \times C_0^{10}}{816} = \frac{56}{816}$ $\quad P(Z|A) = \frac{C_3^5}{816} = \frac{10}{816}$ $\quad P(ZA) = P(Z|A) \times P(A)$

$P(B) = \frac{C_2^8 \times C_1^{10}}{816} = \frac{280}{816}$ $\quad P(Z|B) = \frac{C_3^6}{816} = \frac{20}{816}$ $\quad P(ZB) = P(Z|B) \times P(B)$

$P(C) = \frac{C_1^8 \times C_2^{10}}{816} = \frac{360}{816}$ $\quad P(Z|C) = \frac{C_3^7}{816} = \frac{35}{816}$ $\quad P(ZC) = P(Z|C) \times P(C)$

$P(D) = \frac{C_0^8 \times C_3^{10}}{816} = \frac{120}{816}$ $\quad P(Z|D) = \frac{C_3^8}{816} = \frac{56}{816}$ $\quad P(ZD) = P(Z|D) \times P(D)$

$$P(Z) = P(ZA) + P(ZB) + P(ZC) + P(ZD) = \frac{560 + 5600 + 12600 + 6720}{816 \times 816} = \frac{25480}{665856} \approx 3.827\%$$

Ans: 3.827 %